

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PNG 526 — Past Questions & Materials
by Renance

Course: PNG 526
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 500

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Paper 1 of 2 — PNG 526

Level 500 • Second Semester • PastQuestion

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EFFURUN
COLLEGE OF TECHNOLOGY
DEPARTMENT OF PETROLEUM ENGINEERING
SECOND SEMESTER TEST, 2020/2021 SESSION

PNG 526: RESERVOIR SIMULATION
TIME ALLOWED: 30 MINS
INSTRUCTIONS: ANSWER ALL

1. Write the 2D mathematical equation for modeling a linear, homogenous, and isotropic reservoir.

$$\frac{\partial^2 p}{\partial x^2} + k_y \frac{\partial^2 p}{\partial y^2} = 0$$

2. State three application of 3D mathematical model

- ✓ Used to represent the reservoir in three directions
- ✓ May be dependent on time
- ✓ Represents the reservoir better than 1D and 2D models.
- ✓ Best used for regions far away from the wellbore
- ✓ Used to compute pressures histories.
- ✓ Used to study layered reservoirs.
- ✓ Most difficult to solve

3. Discretize explicitly, the 3D linear partial differential equation with respect to space and time

$$\frac{p_{i,j,k} - 2p_{i,j,k} + p_{i,j,k+1}}{\Delta x^2} + \frac{p_{i,j,k} - 2p_{i,j,k} + p_{i,j,k+1}}{\Delta y^2} = \frac{p_{i,j,k} - p_{i,j,k+1}}{\Delta t}$$

4. State three equilibration data that are used to establish

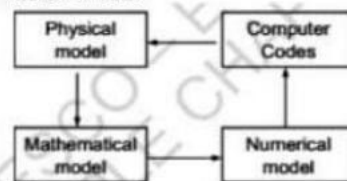
initial reservoir conditions

- Fluid contact depths – oil-water contact (OWC), gas-oil contact (GOC) or gas-water contact
- Capillary pressure at the fluid contacts
- Reservoir pressure at a selected datum depth

5. What is bright spot in seismic interpretation?

Bright spot is referred to as a spot with a local increase of amplitude associated with hydrocarbon accumulations

6. Use a flowchart only, show the process of developing a reservoir simulator



7. State the features of black oil simulators

- It is used to simulate dry gas and black oil reservoirs
- Three phase exists (gas, oil and water)
- Gas may be dissolved in both oil and water
- Does not consider changes in composition of the hydrocarbons as the field is produced

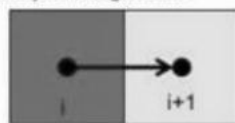
8. List five information that can be obtained from a simulation study

– Development strategy for maximum hydrocarbon

recovery

- Ultimate economic recovery
- Best completion method
- Suitable enhanced recovery scheme for the reservoir & design parameters
- Computer models are also used to complement the physical models

9. Calculate the inter-block permeability and transmissibility from the grid below



$i+1/2$

$$(P_{i+1} - P_i) = \frac{q \mu \beta_o \Delta x}{k_{i+1/2} A}$$

averaging permeability: Harmonic average



$i+1/2$

Pressure drop in green "half block"

$$(P_{i+1/2} - P_i) = \frac{q \mu \beta_o \Delta x}{2k_i A}$$

Pressure drop in yellow "half block"

$$(P_{i+1} - P_{i+1/2}) = \frac{q \mu \beta_o \Delta x}{2k_{i+1} A}$$

Add pressure drops across the green and yellow "half blocks"

$$(P_{i+1/2} - P_i) + (P_{i+1} - P_{i+1/2}) = \frac{q \mu \beta_o \Delta x}{2k_i A} + \frac{q \mu \beta_o \Delta x}{2k_{i+1} A}$$

$$(P_{i+1} - P_i) = \frac{q \mu \beta_o \Delta x}{2A} \left(\frac{1}{k_i} + \frac{1}{k_{i+1}} \right)$$

$$q = \frac{A}{\mu \beta_o \Delta x} \frac{2 \left(\frac{1}{k_i} + \frac{1}{k_{i+1}} \right)^{-1}}{k_{i+1/2}} (P_{i+1} - P_i)$$

Inter-block permeability

$$k_{i+1/2} = 2 \left(\frac{1}{k_i} + \frac{1}{k_{i+1}} \right)^{-1}$$

Inter-block Transmissibility

$$T_{i+1/2} = \frac{k_{i+1/2} A}{\mu \beta_o \Delta x}$$

Solution:

$$V_b = 7758 \times A \times h = 7758 \times 26,700 \times 49 = 10.15 \text{ MMM bbl}$$

1) The initial oil in place is given by:

$$N_i = \frac{V_b \phi (1 - S_{wi})}{B_{oi}}$$

Which yields:

$$N_i = \frac{10.15 \times 10^9 (0.08) (1 - 0.45)}{1.68} \approx 266 \text{ MM STB}$$

2) Oil in place after volumetric depletion to abandonment pressure is given by:

$$N = \frac{V_b \phi (1 - S_w - S_g)}{B_o}$$

Which yields:

$$N_1 = \frac{10.15 \times 10^9 (0.08) (1 - 0.45 - 0.34)}{1.15} \approx 148 \text{ MM STB}$$

3) Oil in place after water invasion at initial reservoir pressure is given by:

$$N = \frac{V_b \phi S_{or}}{B_o}$$

Which yields:

$$N_2 = \frac{10.15 \times 10^9 (0.08) 0.20}{1.68} \approx 97 \text{ MM STB}$$

4) Oil reserve by volumetric depletion

$$(N_i - N_1) = (266 - 148) \times 10^6 = 118 \text{ MM STB}$$

i.e. RF = 118/266 = 44%

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SECOND SEMESTER EXAMINATION, 2020/2021 SESSION

PNG 526: RESERVOIR SIMULATION
TIME ALLOWED: 3HRS UNIT: 3

NOTE: ALL CALCULATIONS SHOULD BE REPORTED IN
FOUR DECIMAL PLACE

SECTION A: ANSWER ANY 3 QUESTIONS FROM
THIS SECTION

QUESTION 1

- a) PVT data derived from laboratory experiments or correlations, are key input parameters for reservoir studies. You are required to generate a table in the format below to list 5 PVT experiments, the type of fluid they can be applied on and three data obtained from each experiment.

Experiment	Type of fluid	Data obtained

- b) State three equilibration data that are used to establish initial reservoir conditions

QUESTION 2

- a) State two parameters each that can be adjusted to achieve a good history match for the following: rock data, fluid data, relative permeability, individual well completion data
- b) Make sketches with a label of bubble point pressure, saturated and undersaturated regions of black oil properties such as

- Formation volume factor for oil, gas, & water respectively with one plot
- Viscosity for oil, gas, & water respectively with one plot
- Solution gas-oil ratio.

QUESTION 3

- (i) You have been contracted by your company to review data affecting STOIP, if your model value is too high/low, what could be five possible sources of errors?
- (ii) State Use a flowchart only, draw a simplified procedure for history matching of:
- Pressure
 - Saturation
- (iii) If the simulation field pressure is too high, what are the possible changes that could be made to match the history pressure
- (iv) In a reservoir simulation model, what are the possible changes that can be made when gas breakthrough is ok, and after breakthrough, the simulation slope is in error

QUESTION 4

- Write the 3D mathematical equation for modeling an Anisotropic reservoir undergoing unsteady state flow.
- State three mathematical model equations that are usually solved in reservoir simulation
- State four deliverables under the following:
 - Seismic review
 - Geologic model
 - BHP Analysis

Paper 2 of 2 — PNG 526

Level 500 • Second Semester • PastQuestion

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SECOND SEMESTER EXAMINATION, 2020/2021 SESSION

PNG 526: RESERVOIR SIMULATION
TIME ALLOWED: 3HRS UNIT: 3

**NOTE: ALL CALCULATIONS SHOULD BE REPORTED IN
FOUR DECIMAL PLACE**

**SECTION A: ANSWER ANY 3 QUESTIONS FROM
THIS SECTION**

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Experiment	Type of fluid	Data obtained
254, 102, 100		

b) State three equilibration data that are used to establish initial reservoir conditions

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QUESTION 4

- i. Write the 3D mathematical equation for modeling an Anisotropic reservoir undergoing unsteady state flow.
- ii. State three mathematical model equations that are usually solved in reservoir simulation
- iii. State four deliverables under the following:
 - a) Seismic review
 - b) Geologic model
 - c) BHP Analysis

SECTION B: ANSWER ANY 2 QUESTIONS FROM THIS SECTION

QUESTION 5

Show that the diffusivity equation for a slightly compressible fluid is given by:

$$\frac{\partial^2 P}{\partial r^2} + \frac{1}{r} \frac{dP}{dr} = \frac{\phi \mu C_t}{k} \frac{\partial P}{\partial t}$$

QUESTION 6

The expression below represents a 1D partial differential equation of a single-phase flow reservoir

$$\frac{0.000264 k_o}{\phi \mu_o \beta_o c_o} \frac{\partial^2 p}{\partial x^2} + \frac{q_o}{B_o} = \frac{\partial p}{\partial t}$$

If the reservoir is discretized into five grid blocks with the following boundary conditions:

- (i) Characterize the associated matrix fully for all the implicit pressures
- (ii) Write down the implicit pressure expression at $i+2, n-1$

QUESTION 7

- (i) Use a flowchart only to describe a reservoir simulation model selection in increasing complexity and expense with respect to system, dimension, geometry and phases
- (ii) Use a flowchart only, describe the process of reservoir simulation study from problem definition to development plan strategies.

PKF 526

2020/2021

Question 1. List 5 PVT Experiments the type of fluids they can be applied to and three data obtained each.

Experiment	Type of fluid	Data Obtained
Saturation Pressure	Black oil & Compositional	Dew point at Reservoir Temperature
Constant Composition Expansion / Flash liberation	Black oil & Compositional	- Relative Volume - Vapor Z factor - Liquid dropout
Differential Liberation / Vaporization	Black oil only	- Vapor Z factor - GOR - Liquid density - Gas gravity - M_L - M_G
Constant Volume Depletion	Compositional only.	- Vapor Z factor - Cumulative fluid produced - Retrograde liquid dropout - Produced vapor composition.
Separate Test	Black oil & Compositional	- FVF (B_o) - GOR

b. State three equilibrium data that are used to establish initial reservoir conditions:

1. Fluid Contact depths
2. Capillary pressure at the fluid contact
3. Reservoir pressure at a selected depth

2. State two parameters each that can be adjusted to achieve a good history match for the following

rock data:

Permeability

Saturation

Porosity

Thickness

Fluid data:

Compressibility

Viscosity

PVT data

Relative permeability:

Relative permeability Curve

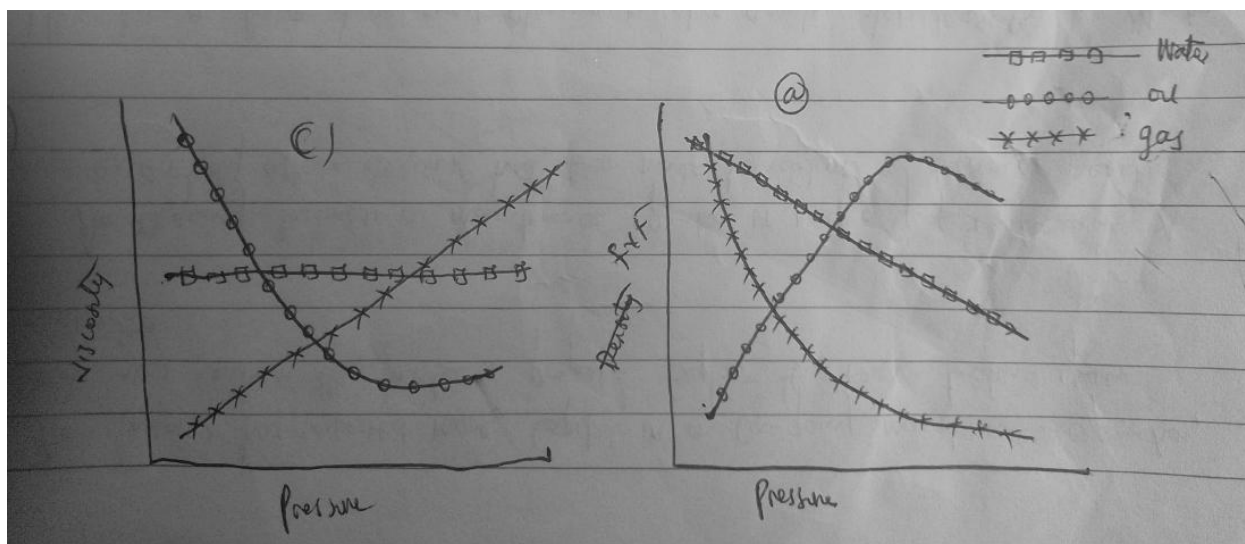
Coreal saturation data

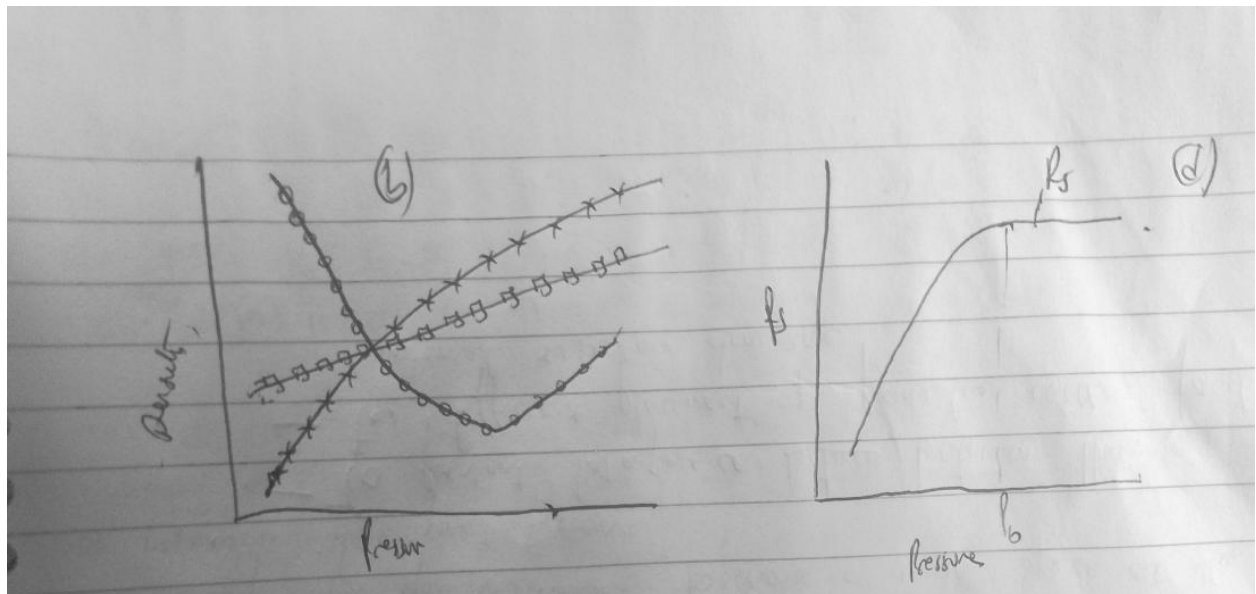
Individual Well Completion data.

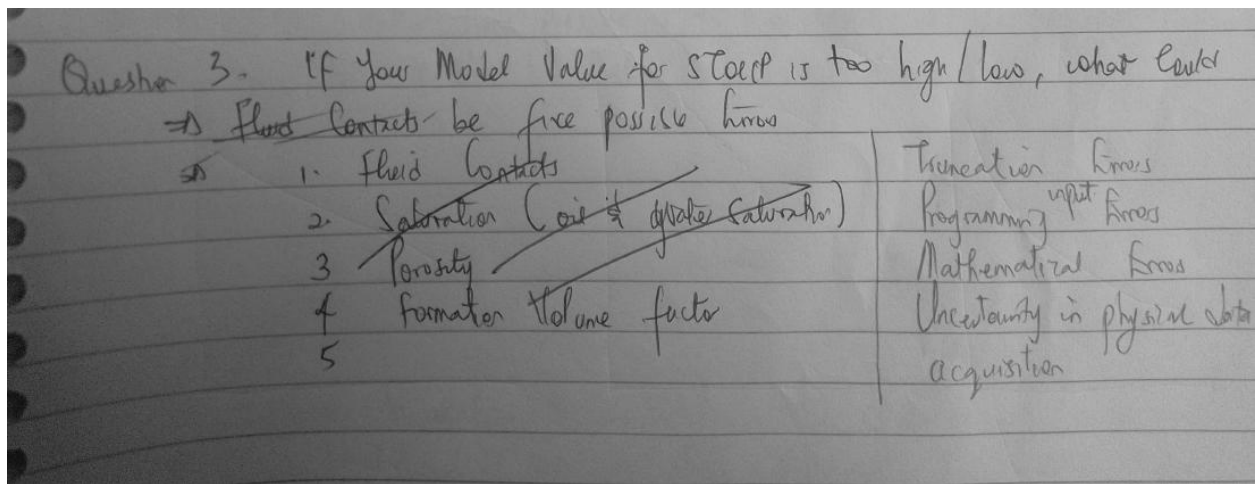
Skin

Flowing wellbore Pressure.

b. Make sketches with a label of bubble point pressure, Saturated and Undersaturated regions of black oil reservoirs

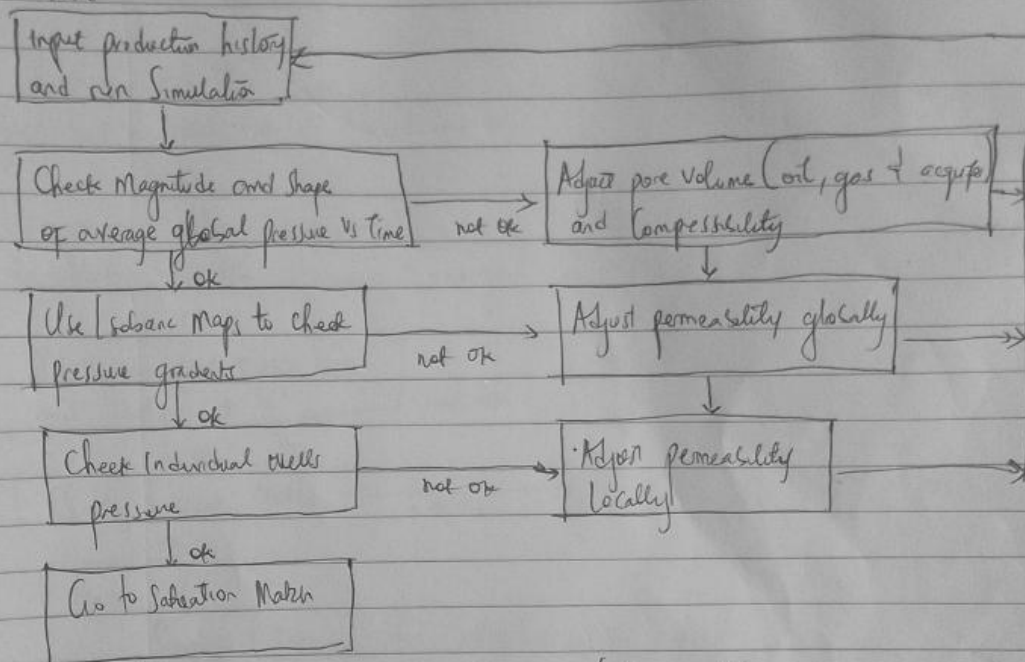






- b) State Using a flatchart Only, draw a simplified procedure for history matching of
- Pressure
 - Saturation

Pressure:



Simplified procedure for pressure history matching

- c. IF the Simulation field pressure is too high what are the possible changes that could be made to match the history possible

Pore Volume

Aquifer Size

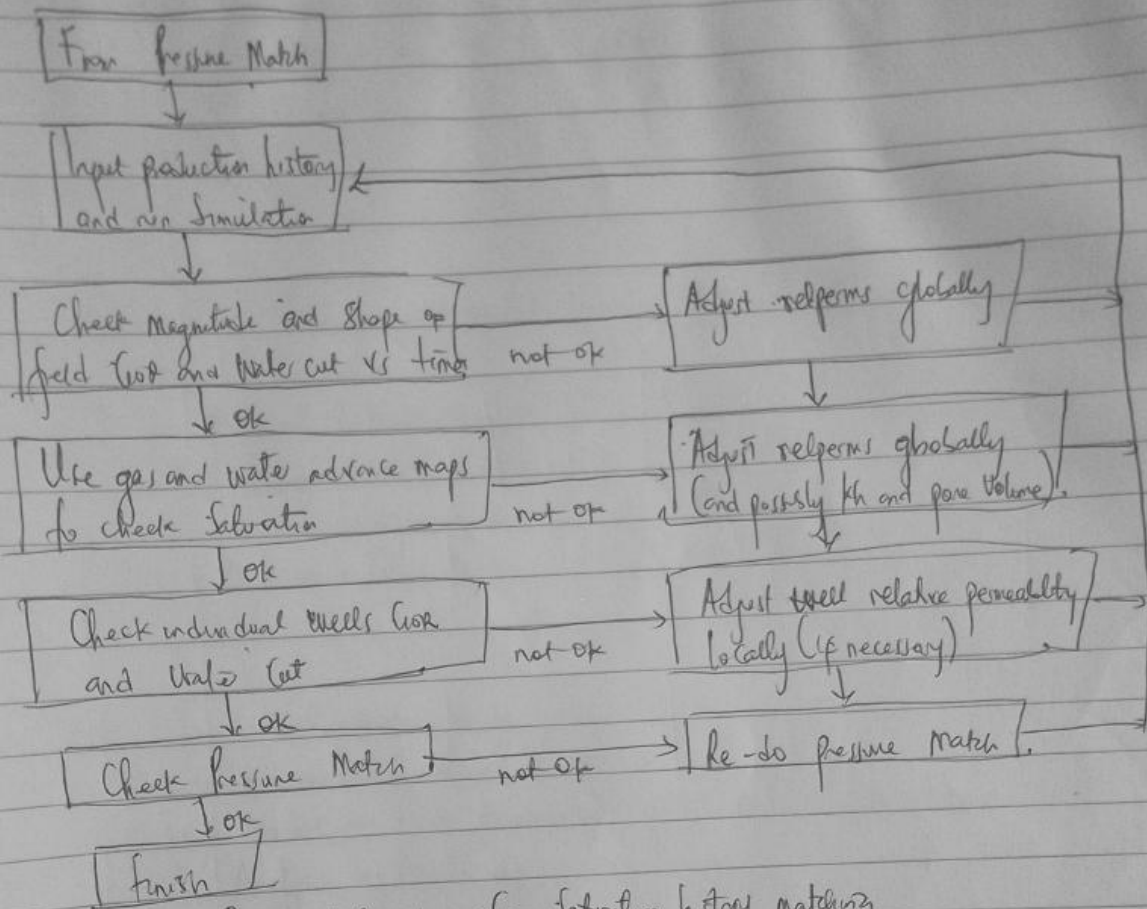
Oil initially in place

— Contacts, So

Energy — Gas Cap Size

- d) In a reservoir Simulation Model, what are the changes that can be made when gas breakthrough is OK, and after breakthrough, the simulation slope is in error.

3b (12)
Salvation



Simplified procedure for saturation history matching

d (12)

$\frac{k_{rg}}{k_{ro}}$ increase
Supply of gas

4. Write the 3D Mathematical equation for Modelling Anisotropic reservoir Undergoing Unsteady flow

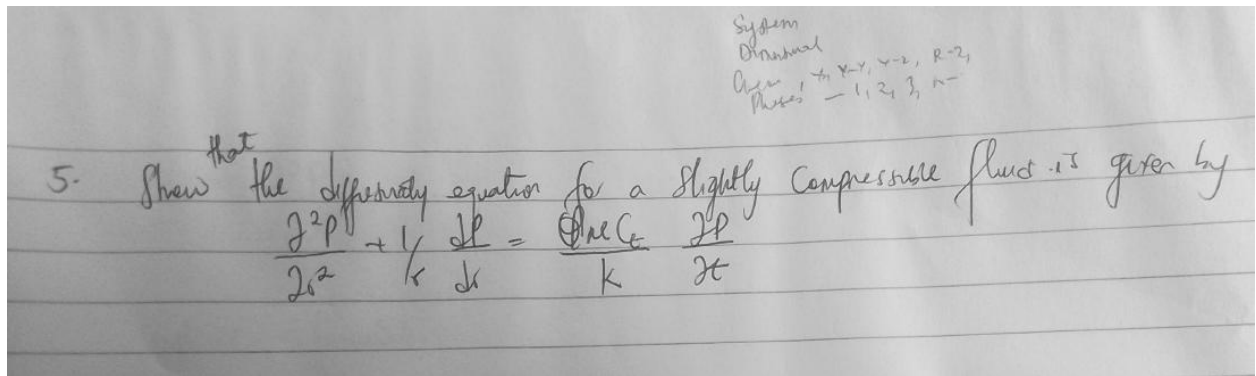
$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} + \frac{\partial^2 p}{\partial z^2} = \frac{\mu \phi C_v}{k} \frac{\partial p}{\partial t}$$

a. State three Mathematical Model equations that are usually solved in reservoir Simulation

- Material Balance Model	Tank Model	3D Model
- Buckley-Leverett Model	1D Model	
- Pressure analysis Model	2D Linear Model	
- Dykstra - Parsons Model	2D Radial Model	

or State four deliverables Under the following

- Section review - Contours depth
- Geologic Model - Rock permeability properties (Porosity, Permeability)
- BHP Analysis - Build up, Draw Down



6.7.3 For slightly Compressible Fluid

The equation below is the diffusivity equation for slightly compressible fluid for a linear flow.

Assumptions

1. Fluid compressibility is very small.
2. Single phase flow
3. Permeability and viscosity are independent of pressure
4. No reaction between fluid and medium
5. Homogenous and isotropic porous medium, and isothermal fluid flow

100 |

Application of Mathematical Techniques for Engineering and Science Practice

6. Steady state condition

$$-\frac{1}{r} \frac{\partial(\rho u r)}{\partial r} = \frac{\partial(\rho \phi)}{\partial t}$$

For liquid

$$\rho = \rho_b e^{C(P-P_b)}$$

$$u = -\frac{k}{\mu} \frac{dP}{dr}$$

By substituting these equation into the diffusivity equation, we have

$$-\frac{1}{r} \frac{\partial}{\partial r} \left[\rho_b e^{C(P-P_b)} \cdot r \cdot \left(-\frac{k}{\mu} \frac{dP}{dr} \right) \right] = \frac{\partial}{\partial t} [\rho_b e^{C(P-P_b)} \phi]$$

Considering the left hand side only

For homogeneous isotropic, $k = \mu$ constant

$$\frac{\rho_b k}{r \mu} \frac{\partial}{\partial r} \left[e^{C(P-P_b)} \cdot r \cdot \frac{dP}{dr} \right]$$

Applying product rule

$$\frac{\rho_b k}{r \mu} \left[e^{C(P-P_b)} \cdot r \cdot \frac{\partial^2 P}{\partial r^2} + e^{C(P-P_b)} \frac{dP}{dr} + r \frac{dP}{dr} e^{C(P-P_b)} C \frac{dP}{dr} \right]$$

$$\rho_b e^{C(P-P_b)} \frac{k}{\mu} \left[\frac{\partial^2 P}{\partial r^2} + \frac{1}{r} \frac{dP}{dr} + C \left(\frac{dP}{dr} \right)^2 \right]$$

Considering the right hand side

By product rule, we have

$$\frac{\partial}{\partial t} [\rho_b e^{C(P-P_b)} \phi] = \left[\rho_b e^{C(P-P_b)} \frac{\partial \phi}{\partial t} + \phi \rho_b e^{C(P-P_b)} C \frac{\partial P}{\partial t} \right]$$

$$\rho_b e^{C(P-P_b)} \left[\frac{\partial \phi}{\partial t} + \phi C \frac{\partial P}{\partial t} \right]$$

Factorized $\phi \frac{\partial P}{\partial t}$ out of the bracket, we have:

$$\rho_b e^{C(P-P_b)} \cdot \phi \frac{\partial P}{\partial t} \left[\frac{1}{\phi} \frac{\partial \phi}{\partial P} + C \right]$$

But

$$C_r = \frac{1}{\phi} \frac{\partial \phi}{\partial P} \quad C_t = C_r + C_f$$

Application of Mathematical Techniques for Engineering and Science

$$\therefore \frac{1}{\phi} \frac{\partial \phi}{\partial P} + C = C_r + C_f = C_t$$

Therefore, the equation is given as:

$$\rho_b e^{C(P-P_b)} \cdot \phi C_t \frac{\partial P}{\partial t}$$

Combining both sides of the equation

$$\rho_b e^{C(P-P_b)} \frac{k}{\mu} \left[\frac{\partial^2 P}{\partial r^2} + \frac{1}{r} \frac{dP}{dr} + C \left(\frac{dP}{dr} \right)^2 \right] = \rho_b e^{C(P-P_b)} \cdot \phi C_t \frac{\partial P}{\partial t}$$

$$\rho_b e^{C(P-P_b)} \frac{k}{\mu} \left[\frac{\partial^2 P}{\partial r^2} + \frac{1}{r} \frac{dP}{dr} + C \left(\frac{dP}{dr} \right)^2 \right] = \rho_b e^{C(P-P_b)} \cdot \phi C_t \frac{\partial P}{\partial t}$$

$$\frac{\partial^2 P}{\partial r^2} + \frac{1}{r} \frac{dP}{dr} + C \left(\frac{dP}{dr} \right)^2 = \frac{\phi \mu C_t}{k} \frac{\partial P}{\partial t}$$

Ignoring the higher degree of the first order derivative wrt r, then

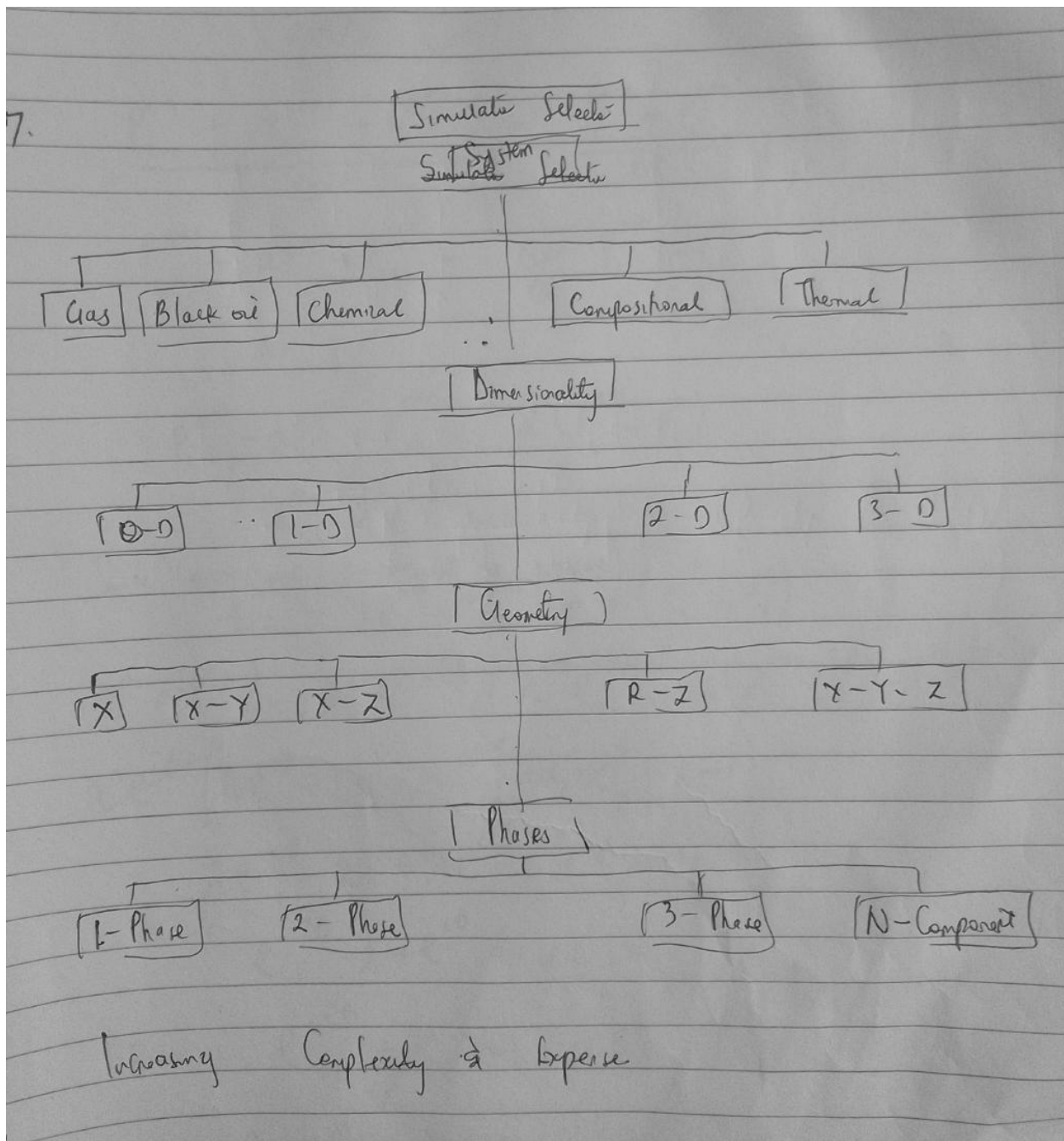
$$\frac{\partial^2 P}{\partial r^2} + \frac{1}{r} \frac{dP}{dr} = \frac{\phi \mu C_t}{k} \frac{\partial P}{\partial t}$$

6.7.4 Diffusivity equation for gases

$$-\frac{1}{r} \frac{\partial(\rho u r)}{\partial r} = \frac{\partial(\rho \phi)}{\partial t}$$

For gas:

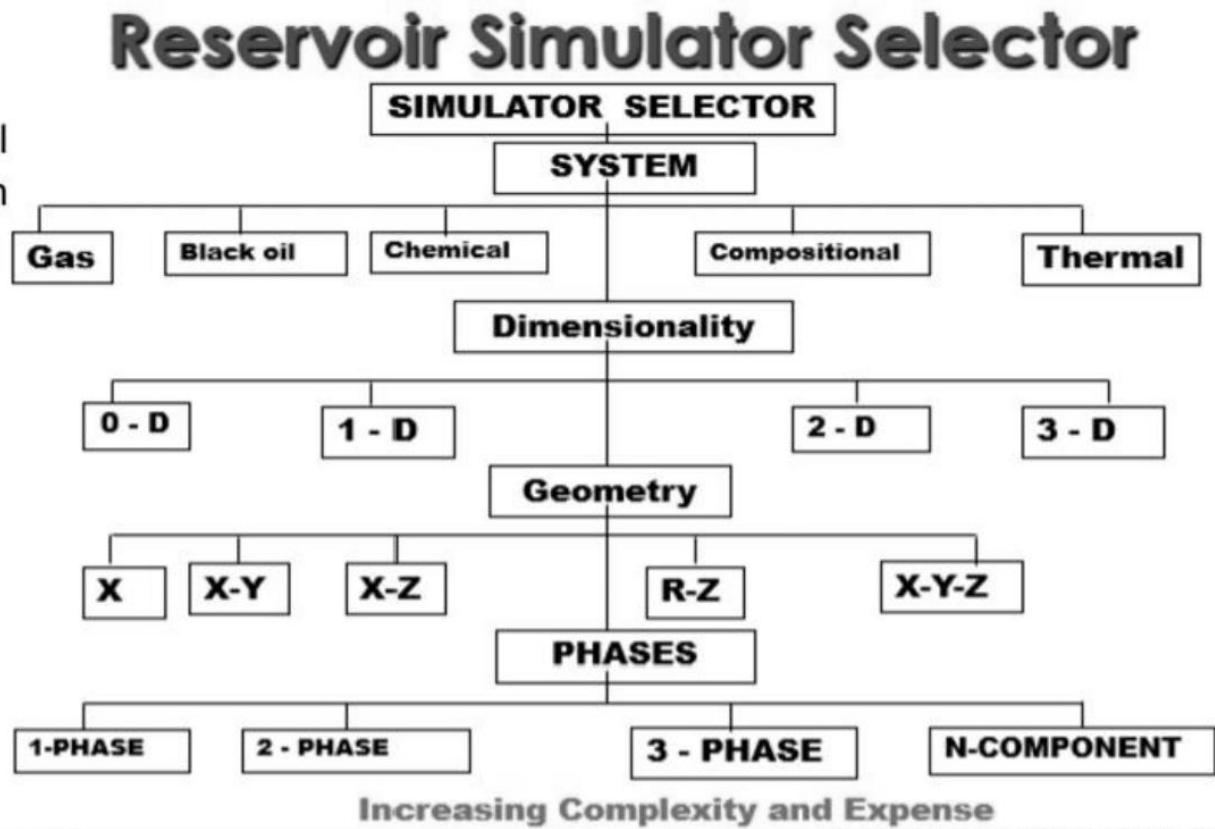
$$\rho = \frac{PM}{zRT}$$



Reservoir Modelling

model and approach

level
on



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DEPARTMENT OF PETROLEUM ENGINEERING
SECOND SEMESTER EXAMINATION, 2019/2020 SESSION

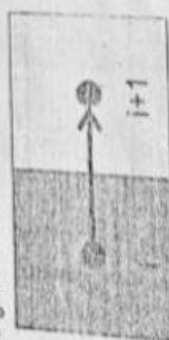
PNG 526: RESERVOIR SIMULATION
TIME ALLOWED: 3 HRS
UNIT: 3
INSTRUCTIONS: ANSWER ALL QUESTIONS IN SECTION
A & ONE IN SECTION B

NOTE: ALL CALCULATIONS SHOULD BE REPORTED IN
FOUR DECIMAL PLACE

SECTION A

QUESTION 1

- State the laws use in reservoir simulation
- Write the governing equations for each of the laws above
- Calculate the inter-block permeability and transmissibility from the grid below

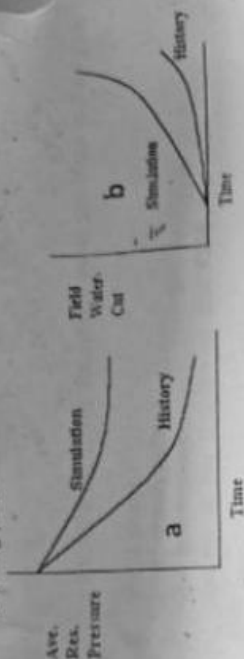


$i+1/2$

- State two deliverables under the following:

- Seismic review
- Geologic model
- BHP Analysis

(V) State the pressure
following plots:



QUESTION 2

The expression below represents a 1D partial differential equation of a single phase flow reservoir

$$\frac{0.000264k_o}{\phi\mu_o\beta_o c_o} \frac{\partial^2 p}{\partial x^2} + \frac{q_o}{B_o} = \frac{\partial p}{\partial t}$$

If the reservoir is discretize into five grid blocks with the following boundary conditions: inner boundary condition: $P(0, t) = a$ and outer boundary condition $P(L, t) = 2b$

- Derive a procedure for computing all the pressures explicitly
- Derive an associated matrix for all the implicit pressures
- Characterize the associated matrix fully
- Write down the explicit pressure expression at $i+2, n-1$

QUESTION 3

Determine the stability of the implicit formulation for a 1D heterogeneous PDE given below

$$\frac{\partial^2 p}{\partial x^2} = \frac{\partial p}{\partial t}$$

Sub Success

QUESTION 4

- (i) You have been contacted by your company to review data affecting STOLIP, if your model value is too high/low, what could be five possible sources of errors?
- (ii) State the procedure for matching
 - a) Pressure
 - b) Saturation
- (iii) If the simulation field pressure is too high, what are the possible changes that could be made to match the history pressure
- (iv) In a reservoir simulation model, what are the possible changes that can be made when gas breakthrough is ok, and after breakthrough, the simulation slope is in error

SECTION B

QUESTION 5

- i. Make sketches with a label of bubble point pressure, saturated and undersaturated regions of black oil properties such as
 - a. Formation volume factor for oil, gas & water respectively with one plot
 - b. Oil, gas and water densities in one plot
 - c. Viscosity for oil, gas, & water respectively with one plot
 - d. Solution gas-oil ratio
- ii. Despite increasing reliance on simulators, state three key limitations
- iii. State the three ways grid blocks can be identified and ordered. Also, identify the blocks in a 4x3 reservoir, using any method of your choice.

QUESTION 6

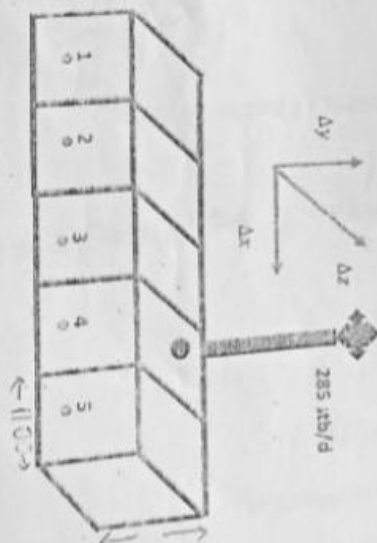
Assume the following for the discretized equal grid block in the figure below:

- i. A single phase
- ii. Neglect gravity term
- iii. Water and rock are incompressible
- iv. Pressure at the boundaries are 4200 psia and 1150 psia respectively

Find the pressure distribution in the grid blocks using explicit formulation.

Data given:

$$\begin{aligned} \phi &= 34.5\%, k_r = 200 \text{ mD}, \mu_o = 0.86 \text{ cp}, \beta_o = 1.23 \frac{\text{rb}}{\text{stb}}, \beta_c \\ &= 0.001127, \gamma_o = 0.21584, g = 32.174 \text{ ft/sec}^2, \Delta x \\ &= 1100 \text{ ft}, \Delta y = 50 \text{ ft}, \Delta z = 600 \text{ ft} \end{aligned}$$



$$2000 - 1000 = 1000$$

$$-2000 + 2000 = 0$$

2019/2020 Session

1) i) State the laws used in reservoir Engineering

- Law of Conservation of Momentum
- Law of Conservation of Mass
- Law of Conservation of Energy

ii) Write the governing equation

Law of Conservation of ~~Momentum~~ Mass

$$\left[\int_{V_0} \rho_0 \frac{\partial}{\partial t} dx + \rho_{in} A \Delta x \right] - \left[\int_{V_0} \rho_0 \frac{\partial}{\partial t} dx + \rho_{out} A \Delta x \right] = \text{Mass Storage}$$

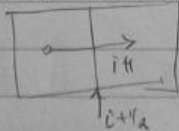
Law of Conservation of ~~Mass~~ Momentum

$$u_i = - \frac{k k_{ri}}{\mu_i} (\nabla p_i + \rho_i \nabla z)$$

Law of Conservation of Energy

$$\text{Salivation } \frac{\partial}{\partial t} \left(\frac{\phi b_0}{b_0} \right) = \frac{\partial}{\partial x} \left[\frac{k k_{ro}}{\mu_0 b_0} \left(\frac{\partial p}{\partial x} - \rho_0 g \frac{\partial z}{\partial x} \right) \right]$$

iii) Calculate the inter-block Permeability and Transmissivity from the graph below



$$\text{recall } q = \frac{k A \Delta p}{\mu B \Delta x}$$

$$\frac{\partial p}{\partial x} = \frac{p_{i+1/2} - p_e}{\Delta x}$$

$$q = \frac{k_{i+1/2} A (p_{i+1/2} - p_e)}{\mu B \Delta x}$$

$$p_{i+1/2} - p_e = \frac{q \mu B \Delta x}{k_{i+1/2} A}$$

For the first block

$$\frac{2p}{2x} = \frac{P_{i+1/2} - P_i}{\Delta x^2}$$

$$P_{i+1/2} - P_i = \frac{q \mu B_w \Delta x}{2 k_i A}$$

For the second block

$$P_{i+1} - P_{i+1/2} = \frac{q \mu B_w \Delta x}{2 k_{i+1} A}$$

Add pressure across both blocks

$$(P_{i+1/2} - P_i) + (P_{i+1} - P_{i+1/2}) = \frac{q \mu B_w \Delta x}{2 k_i A} + \frac{q \mu B_w \Delta x}{2 k_{i+1} A}$$

$$(P_{i+1} - P_i) = \frac{q \mu B_w \Delta x}{2 A} \left(\frac{1}{k_i} + \frac{1}{k_{i+1}} \right)$$

$$q = \frac{2 A \left(\frac{1}{k_i} + \frac{1}{k_{i+1}} \right)^{-1} (P_{i+1} - P_i)}{\mu B_w \Delta x k_{i+1/2}}$$

Inter-block permeability

$$k_{i+1/2} = 2 \left(\frac{1}{k_i} + \frac{1}{k_{i+1}} \right)^{-1}$$

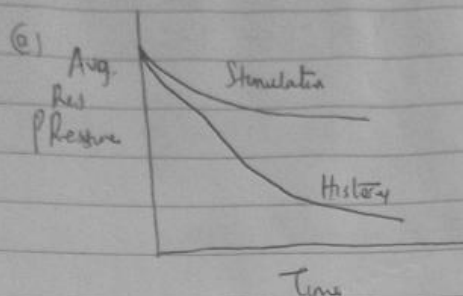
Inter-block transmissibility

$$T_{i+1/2} = \frac{k_{i+1/2} A}{\mu B_w \Delta x}$$

iv. State two deliverables under the following.

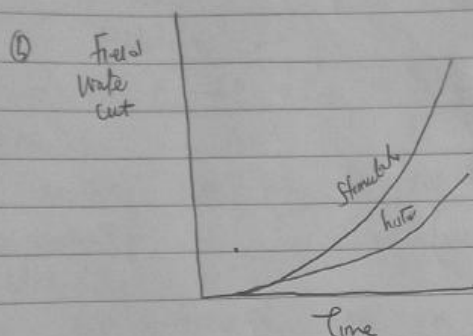
	Deliverables
a. Seismic Review	Identification of HC accumulation Determination of Fluid Water Contact
b. Geologic Model	Reservoir Continuity Stratigraphy, Volumetric Map, Barrier Map
c. BHP Analysis	Production rate of reservoir influx performance Prediction

14 State the possible changes that could be done to match the following plots



Possible Changes

- Pore Volume
- Aquifer Size
- OIP (fluid contacts, S_o)
- Gas cap size



Possible Changes

- k_{rw}/k_{ro} ratio decrease
- Aquifer Size

Q2.

$$\frac{0.000264k_o}{\phi M_o B_o C_o} \frac{\partial^2 P}{\partial x^2} + \frac{q_o}{B_o} = \frac{\partial P}{\partial t} \quad P(C, t) = 9$$

$$P(L, t) = 26$$

Reservoir is discretized into five blocks

c) Derive a procedure for computing all the pressures explicitly

$$\frac{0.000264k_o}{\phi M_o B_o C_o} \left(\frac{\partial^2 P}{\partial x^2} \right)_i + \frac{q_o}{B_o} = \left(\frac{\partial P}{\partial t} \right)_i$$

$$\frac{0.000264k_o}{\phi M_o B_o C_o} \frac{P_{i-1}^n - 2P_i^n + P_{i+1}^n}{\Delta x^2} + \frac{q_o}{B_o} = \frac{P_i^{n+1} - P_i^n}{\Delta t}$$

multiply through by Δt

$$\frac{0.000264k_o}{\phi M_o B_o C_o} \frac{\Delta t}{\Delta x^2} \frac{P_{i-1}^n - 2P_i^n + P_{i+1}^n}{b} + \frac{\Delta t q_o}{B_o} = P_i^{n+1} - P_i^n$$

let $\frac{0.000264k_o}{\phi M_o B_o C_o} \frac{\Delta t}{\Delta x^2} = \alpha$, $\frac{\Delta t q_o}{B_o} = \beta$

$$\begin{aligned} \alpha(P_{i-1}^n - 2P_i^n + P_{i+1}^n) + \beta &= P_i^{n+1} - P_i^n \\ \alpha P_{i-1}^n - 2\alpha P_i^n + \alpha P_{i+1}^n + \beta &= P_i^{n+1} - P_i^n \\ \alpha P_{i-1}^n - 2\alpha P_i^n + \alpha P_{i+1}^n + P_i^n &= P_i^{n+1} - \beta \end{aligned}$$

$$\alpha P_{i-1}^n - (2\alpha - 1)P_i^n + \alpha P_{i+1}^n = P_i^{n+1} - \beta \dots$$

ii) Derive an associated Matrix for all the implicit pressures

$$\frac{0.00026k_0}{\Phi M_0 B_0 C_0} \left(\frac{\partial^2 p}{\partial x^2} \right)_i^{n+1} + \frac{q_0}{B_0} = \left(\frac{\partial p}{\partial t} \right)_i^{n+1}$$

$$\frac{0.00026k_0}{\Phi M_0 B_0 C_0} \frac{P_{i-1}^{n+1} - 2P_i^{n+1} + P_{i+1}^{n+1}}{\Delta x^2} + \frac{q_0}{B_0} = \frac{P_i^{n+1} - P_i^n}{\Delta t}$$

Multiply both sides by Δt

$$\frac{0.00026k_0}{\Phi M_0 B_0 C_0} \frac{\Delta t}{\Delta x^2} \cdot P_{i-1}^{n+1} - 2P_i^{n+1} + P_{i+1}^{n+1} + \Delta t \frac{q_0}{B_0} = P_i^{n+1} - P_i^n$$

Let $\frac{0.00026k_0}{\Phi M_0 B_0 C_0} \frac{\Delta t}{\Delta x^2} = \alpha$, $\Delta t \frac{q_0}{B_0} = \beta$

$$\begin{aligned} \alpha(P_{i-1}^{n+1} - 2P_i^{n+1} + P_{i+1}^{n+1}) + \beta &= P_i^{n+1} - P_i^n \\ \alpha P_{i-1}^{n+1} - 2\alpha P_i^{n+1} + \alpha P_{i+1}^{n+1} + \beta &= P_i^{n+1} - P_i^n \\ \alpha P_{i-1}^{n+1} - 2\alpha P_i^{n+1} + \alpha P_{i+1}^{n+1} - P_i^{n+1} &= -P_i^n - \beta \\ \alpha P_{i-1}^{n+1} - (2\alpha + 1)P_i^{n+1} + \alpha P_{i+1}^{n+1} &= -P_i^n - \beta \end{aligned}$$

When $i=1$

$$\alpha P_0^{n+1} - (2\alpha + 1)P_1^{n+1} + \alpha P_2^{n+1} = -P_1^n - \beta$$

When $i=2$

$$\alpha P_1^{n+1} - (2\alpha + 1)P_2^{n+1} + \alpha P_3^{n+1} = -P_2^n - \beta$$

Since $P_0^{n+1} = P(0, t) = a$

When $i=3$

$$\alpha P_2^{n+1} - (2\alpha + 1)P_3^{n+1} + \alpha P_4^{n+1} = -P_3^n - \beta$$

When $i=4$

$$\alpha P_3^{n+1} - (2\alpha + 1)P_4^{n+1} + \alpha P_5^{n+1} = -P_4^n - \beta$$

$$\alpha P_3^{n+1} - (2\alpha + 1)P_4^{n+1} + 2\alpha b = -P_4^n - \beta$$

recall $P(4, t) = 25 = P_5$

$$\alpha P_2^{n+1} - (2\alpha + 1)P_4^{n+1} = -P_4^n - \beta - 2\alpha b$$

Associated Matrix

$$\begin{bmatrix} -\alpha(2\alpha+1) & \alpha & 0 & 0 \\ \alpha & -(2\alpha+1) & \alpha & 0 \\ 0 & \alpha & -(2\alpha+1) & \alpha \\ 0 & 0 & \alpha & -(2\alpha+1) \end{bmatrix} \begin{bmatrix} -P_1^n - \beta - \alpha a \\ -P_2^n - \beta \\ -P_3^n - \beta \\ -P_4^n - \beta - 2\alpha b \end{bmatrix}$$

ii) It is a tri-diagonal Matrix.

iv) at $i+2, n+1$:

Sub

Let $n = n-1, i = i+2$

$$\alpha P_{(i+2)+1}^{(n-1)+1} - (2\alpha+1)P_{(i+2)}^{(n-1)+1} + \alpha P_{(i+2)+1}^{(n-1)+1} = -P_{(i+2)}^{(n-1)} - \beta$$

$$\alpha P_{i+1}^n - (2\alpha+1)P_{i+2}^n + \alpha P_{i+3}^n = -P_{i+2}^{n-1} - \beta$$

3) Determine the stability of the implicit formulation for a 1D heterogeneous PDE given below

$$\frac{\partial^2 p}{\partial x^2} = \frac{\partial p}{\partial t}$$

Implicit Formulation

$$\left(\frac{\partial^2 p}{\partial x^2}\right)^{n+1} = \left(\frac{\partial p}{\partial t}\right)_i^{n+1}$$

$$\frac{P_{i-1}^{n+1} - 2P_i^{n+1} + P_{i+1}^{n+1}}{\Delta x^2} = \frac{P_i^{n+1} - P_i^n}{\Delta t}$$

$$\frac{\Delta t}{\Delta x^2} (P_{i-1}^{n+1} - 2P_i^{n+1} + P_{i+1}^{n+1}) = P_i^{n+1} - P_i^n$$

$$\text{Let } \frac{\Delta t}{\Delta x^2} = \alpha$$

$$\alpha (P_{i-1}^{n+1} - 2P_i^{n+1} + P_{i+1}^{n+1}) = P_i^{n+1} - P_i^n$$

$$\alpha p_{i-1}^{n+1} - 2\alpha p_i^{n+1} + \alpha p_{i+1}^{n+1} = p_i^{n+1} - p_i^n$$

$$\alpha p_{i-1}^{n+1} - 2\alpha p_i^{n+1} + \alpha p_{i+1}^{n+1} - p_i^{n+1} = -p_i^n$$

$$\alpha p_{i-1}^{n+1} - (2\alpha + 1)p_i^{n+1} + \alpha p_{i+1}^{n+1} = -p_i^n$$

$$\left(\frac{\partial^2 p}{\partial x^2}\right)_i^{n+1} = \left(\frac{\partial^2 p}{\partial x^2}\right)_i^n$$

$$\frac{p_{i-1}^{n+1} - 2p_i^{n+1} + p_{i+1}^{n+1}}{\Delta x^2} = \frac{p_i^{n+1} - p_i^n}{\Delta t}$$

$$p_{i-1}^{n+1} - 2p_i^{n+1} + p_{i+1}^{n+1} = \frac{\Delta t}{\Delta x^2} \Delta x^2 \frac{p_i^{n+1} - p_i^n}{\Delta t}$$

$$\frac{\Delta x^2}{\Delta t} = \alpha$$

$$p_{i-1}^{n+1} - 2p_i^{n+1} + p_{i+1}^{n+1} = \alpha (p_i^{n+1} - p_i^n)$$

$$e_i^n = \lambda^n e^{i\beta i}$$

$$\lambda^n e^{i\beta(i-1)} - 2\lambda^n e^{i\beta i} + \lambda^n e^{i\beta(i+1)} = \alpha (\lambda^n e^{i\beta i} - \lambda^n e^{i\beta i})$$

$$\lambda^n e^{i\beta i} [\lambda e^{-i\beta} - 2 + \lambda e^{i\beta}] = \lambda^n e^{i\beta i} \alpha (\lambda - 1)$$

$$\lambda e^{-i\beta} - 2 + \lambda e^{i\beta} = \alpha (\lambda - 1)$$

$$e^{-i\beta} - 2 + e^{i\beta} = \alpha \left(1 - \frac{1}{\lambda}\right)$$

From trig. $e^{-i\beta} + e^{i\beta} = 2\cos\beta$

$$2\cos\beta - 2 = \alpha \left(1 - \frac{1}{\lambda}\right)$$

$$\frac{2\cos\beta - 2}{\alpha} = 1 - \frac{1}{\lambda}$$

$$\frac{1}{\lambda} = 1 - \frac{2\cos\beta - 2}{\alpha}$$

$$\lambda = \frac{1}{1 - \frac{2\cos\beta - 2}{\alpha}} = \frac{1}{1 + \frac{2}{\alpha}(1 - \cos\beta)}$$

$$\lambda = \frac{1}{1 + \frac{2}{\alpha}(1 - \cos \beta)}$$

$\alpha = \text{Constant}$

$\cos \beta$	λ
0	< 1
-1	< 1
1	$= 1$
0.1	< 1
0.01	< 1
0.001	< 1

$\therefore$ It is Unconditionally Stable

Q4.

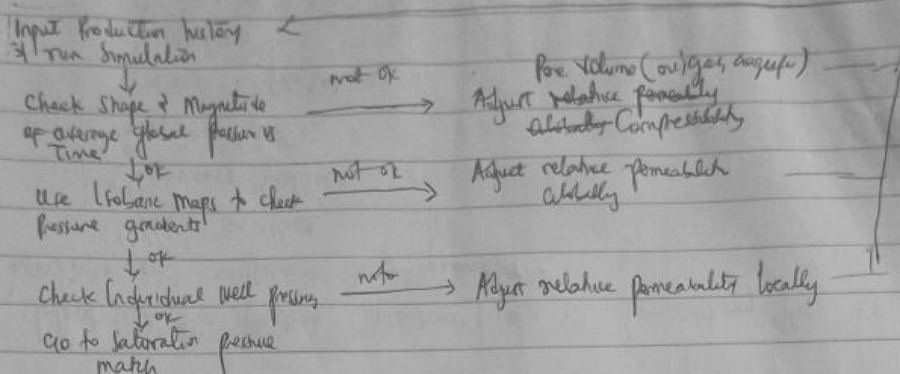
- c). Truncation Error
 Programming input Error
 Uncertainty in Physical Data acquisition
 Mathematical Error
 Model Calibration Error

iii) IF the Simulation field pressure is too high, what are the possible changes that could be made to match the history pressure

Pore Volume
 Aquifer (Size)
 Out initially in place
 - Contacts, So
 Energy - Gas Cap Size

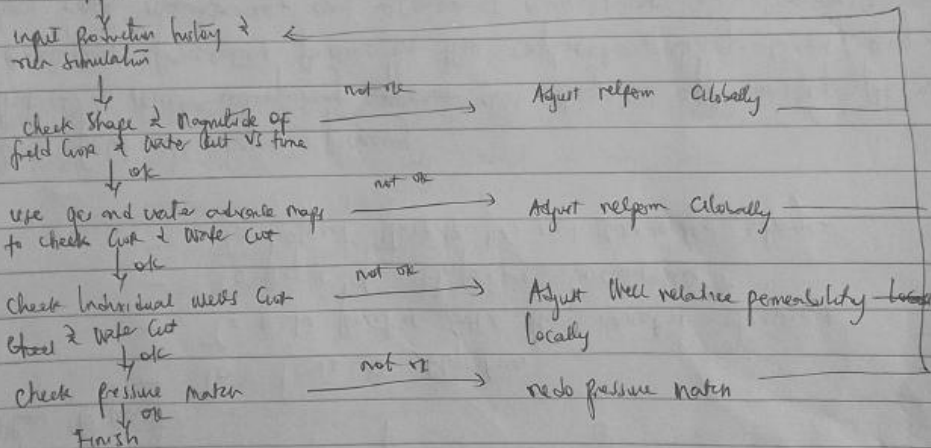
iv) What are the possible changes that can be made when gas breakthrough is OK and after breakthrough, the simulation slope is in Error
 $\frac{k_{rg}}{k_m}$ ratio increase
 Supply of gas

i) Pressure

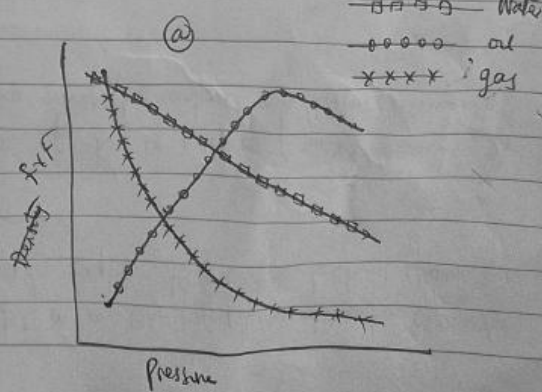
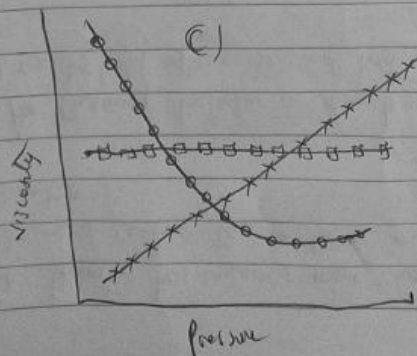


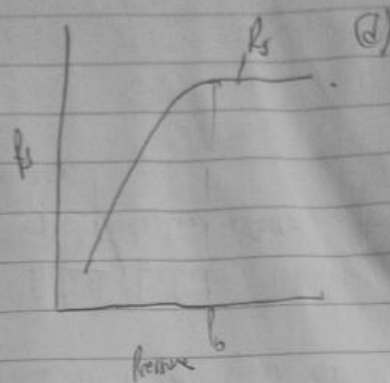
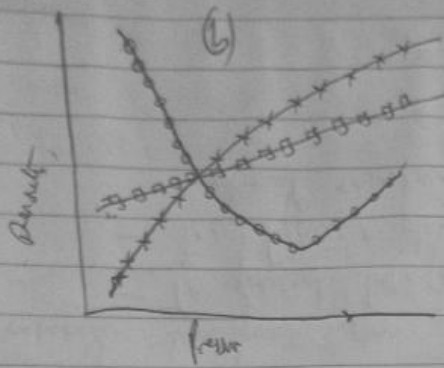
Saturation

From Pressure Match



5i)





iii

State three key limitations on Simulators

- They cannot be used to determine the physics of the problem
- Quality of simulation results depend on quality of the input
- Simulators are just tools and cannot replace good engineering judgement

iv

ii) State the three ways gnd blocks can be identified and ordered - Also identify the blocks in a 4,3 reservoir, using any method of your choice

- Reservoir representation
- Engineering notation
- Natural Ordering of blocks

